BamQuery: a new proteogenomic tool to explore the immunopeptidome

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Introduction

Recent advances in proteogenomics have facilitated deciphering the composition of immunopeptidome presentation, paving the road for the development of immunotherapies.

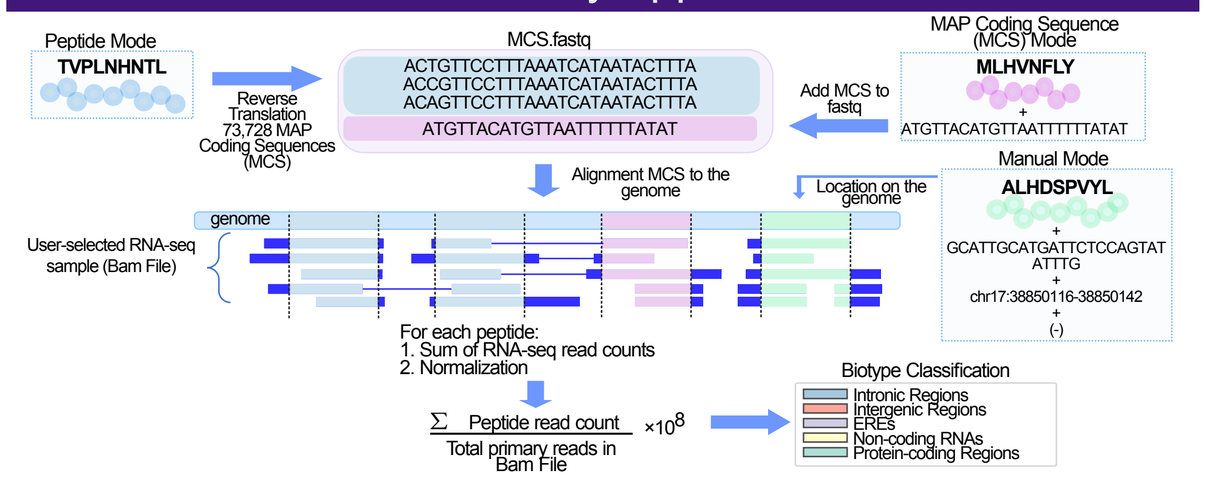
However, the identification of actionable targets of tumor antigens (TAs) has posed three challenges:

1. The attribution of an exact RNA expression to each MHC-I-associated peptide (MAP).
2. Attributing a biotype to MAPs. Due to the multiplicity of genomic regions capable of generating the same MAP, and possibly having different biotypes, a MAP could be mislabeled.
3. Prioritize TAs. TAs must be immunogenic and specifically expressed (or overexpressed) by malignant cells.

To address these challenges, we present BamQuery, an annotation-independent tool that allows comprehensive RNA expression attribution to any MAP in any RNA-seq dataset of interest by considering all genomic regions likely to generate a given MAP.

BamQuery facilitates comparing MAP RNA expression between tumor and normal samples to evaluate TA's specificity. In addition, we show that evaluation of MAPs RNA expression in medullary thymic epithelial cells (mTECs) can be used as an indicator of immunogenicity probability.

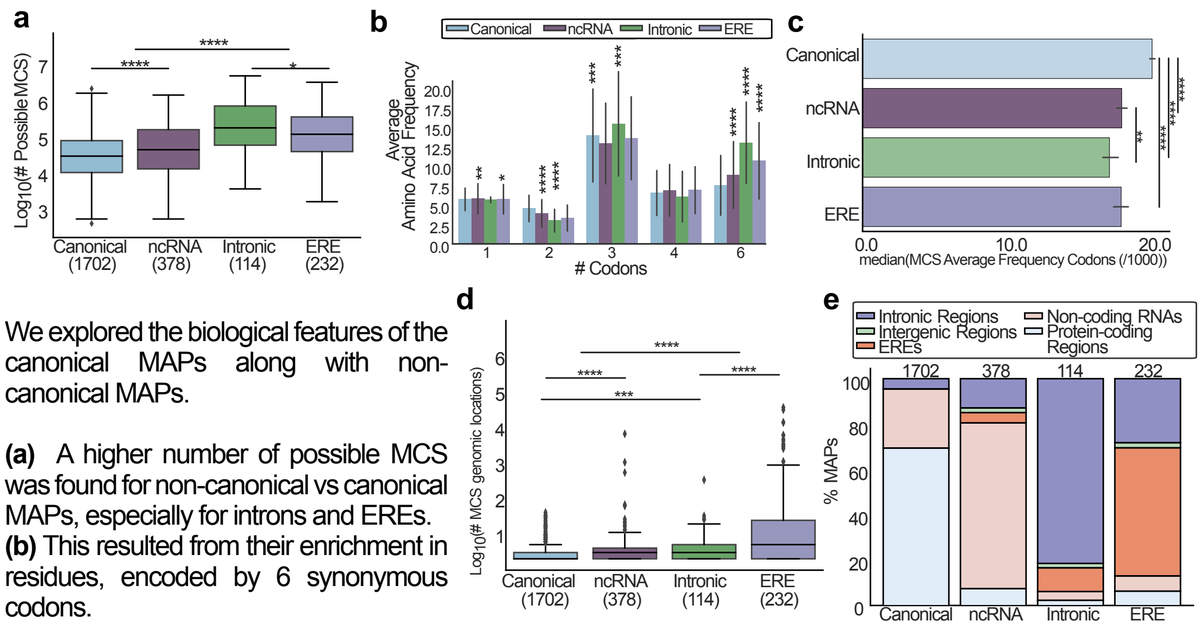
BamQuery Approach



BamQuery work on bam files in five steps : (1) reverse-translate each MAP into all possible MCS; (2) map MCS to the genome (including SNPs) to identify those having perfect matches with the reference and attribute them a genomic location; (3) count the primary RNA-seq reads encompassing exactly the MCS at its respective location, sum read counts of each MAP across locations; (4) normalize the read count of each MAP by the total primary read count of the sample and multiply by 1×10^8 to yield read-per-hundred-million (RPHM) numbers and (5) attribute biotypes to MAPs based on the reference annotations overlapping the various expressed (RPHM>0) regions.

Results

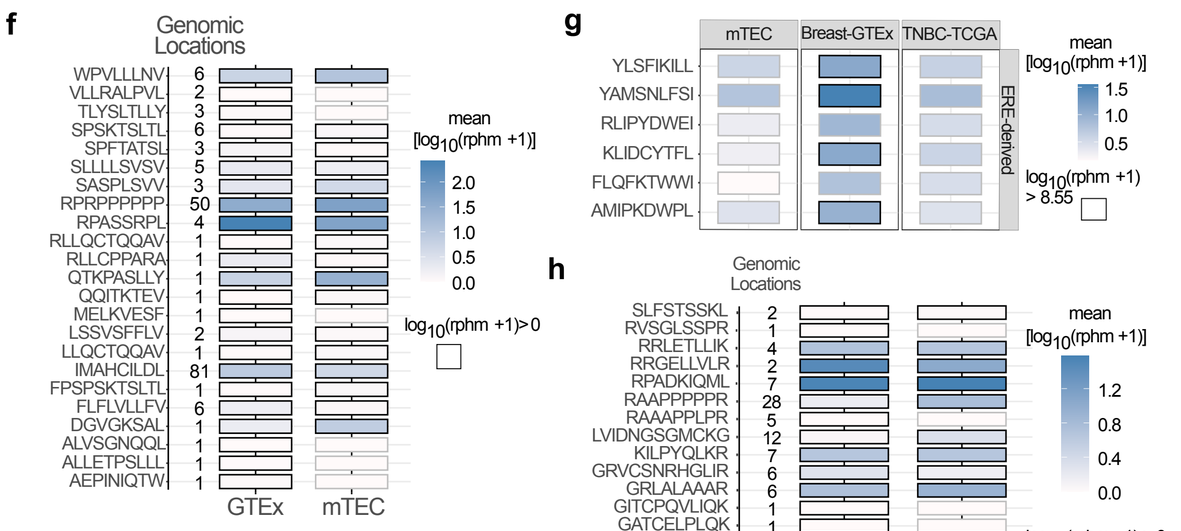
BamQuery provides new insights into immunopeptidome biology



We explored the biological features of the canonical MAPs along with non-canonical MAPs.

(a) A higher number of possible MCS was found for non-canonical vs canonical MAPs, especially for introns and EREs.
(b) This resulted from their enrichment in residues, encoded by 6 synonymous codons.
(c) Given the link between MAP generation and greater frequency of rare codons[1], we observed that canonical MCS codons presented a greater frequency than ncMCS ones. Suggesting that DRiPs contribute more to the generation of ncMAPs than to canonical ones.
(d) Greater # of possible MCS would not automatically result in greater (and proportional) # of genomic regions able to code for a given MAP. Canonical MAPs tended to derive from a single genomic location while ncMAPs could derive from multiple regions.
(e) Given the multiplicity of possible regions of origin, ~26% canonical MAPs most likely to derive from ncRNAs. ncRNA and intron MAPs were predicted to belong mainly from their identified biotype (73 and 81%), only 56% of ERE-derived MAPs were estimated to derive from EREs. 6% of them could derive from canonical regions (5% in-frame). Altogether, these data show that many published MAPs could be mislabeled, either as canonical or non-canonical.

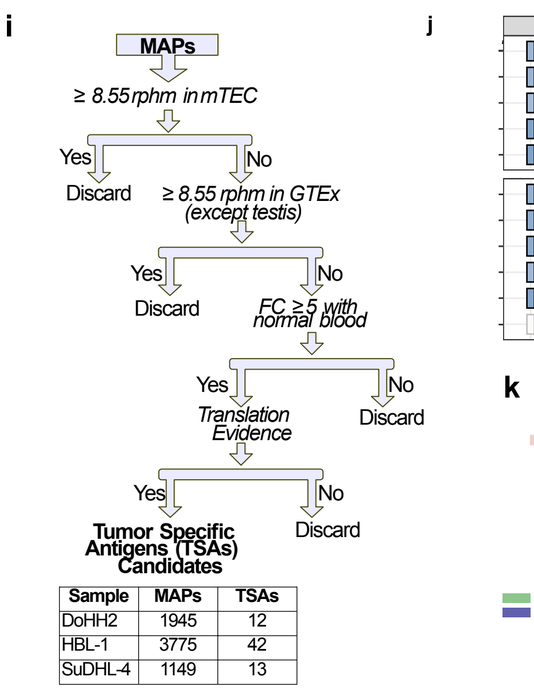
BamQuery discovers underestimated expression in healthy tissues



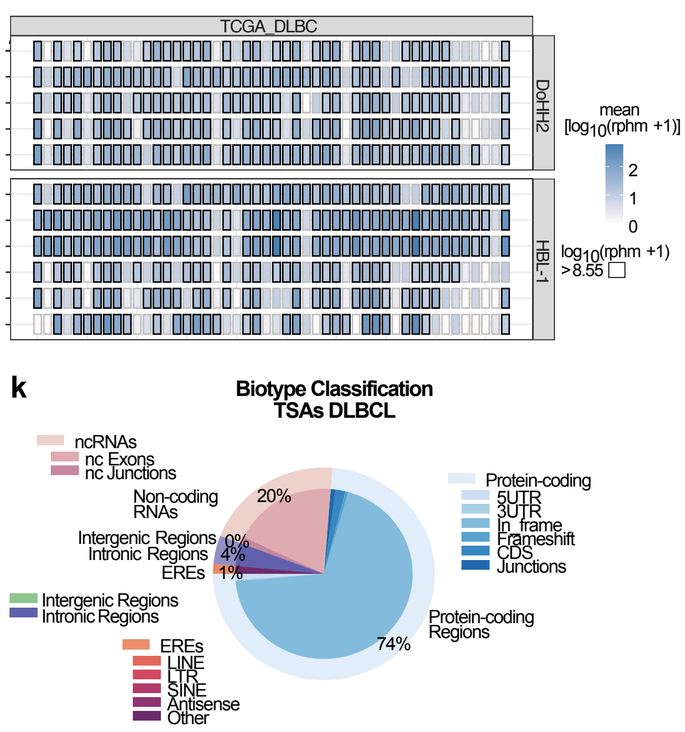
We sought to challenge MAPs previously reported as mutated TAs, ERE-derived TSAs and proteasomal splicing MAPs.

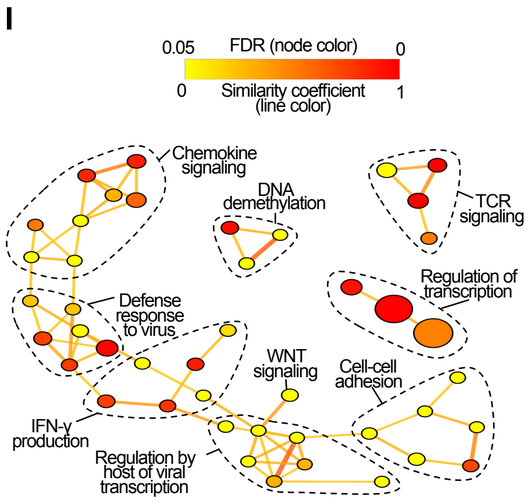
(f) From 323 mutated TAs [2,3,4], 23 (7%) were highly expressed in normal tissues with an average of 8 non-mutated genomic locations perfectly matching their MCS.
(g) For the 6 ERE-derived MAPs reported as TAs in triple negative breast cancer[5], we found elevated expression in normal breast samples, even at greater levels than in cancer samples.
(h) 15 MAPs allegedly deriving from proteasomal splicing[6] could be generated by expressed regions, suggesting a possible misclassification of these peptides.
These results highlight the need to consider all genomic locations capable of generating a given MAP when measuring RNA expression and demonstrate that BamQuery is crucial for attributing comprehensive RNA expression to MAPs, suggesting that it could help discriminate target-safe MAPs.

Discovery of tumor specific antigens in diffuse large B-cell lymphoma



Sample	MAPs	TSAs
DoHH2	1945	12
HBL-1	3775	42
SuDHL-4	1149	13

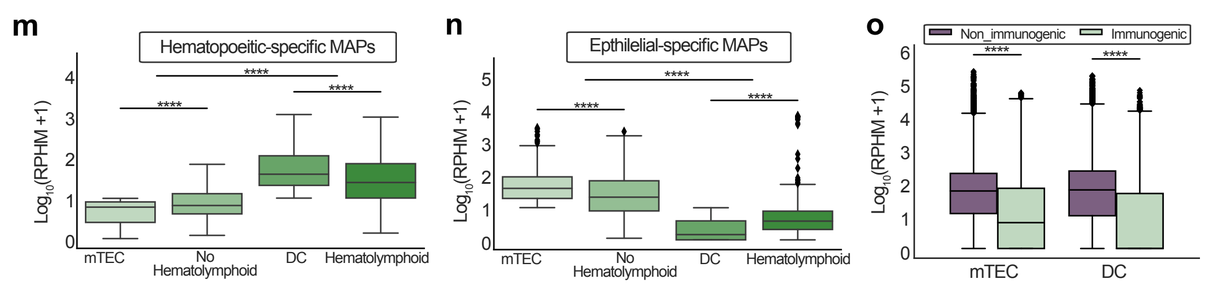




Using BamQuery we identify tumor-specific antigens (TSAs) from raw immunopeptidomic data from 3 DLBCL[7].

(i) Sequential searches with BamQuery on different RNA-seq datasets to filter out uninteresting MAPs, 67 MAPs were flagged as TSA.
(j) Among them, 11 were promising as they were highly shared between DLBCL patients.
(k) Most TSAs derived from protein-coding regions of the genome, as only ~25% of them derived from nc RNA (20%), EREs (1%) and intronic (4%) regions.
(l) Expression of TSAs correlated with TCR signaling and other pro-inflammatory responses in DLBCL patients supporting the biological value of TAs discovered with BamQuery.

BamQuery: online tool to facilitate TAs prioritization



We implemented an online portal (http://bamquery.iric.ca/) to perform analyses on user-defined lists of MAPs. We included queries in mTECs and DCs as proxy of tumor-specificity and immunogenicity, as mTECs share more transcriptomic features with epithelial than hematopoietic cells. We included DCs as they are assumed to exert a non-redundant role in central tolerance establishment with mTECs.

(m,n) Hematopoietic and epithelial MAPs lowly expressed in mTECs and DCs, respectively. This evidenced dramatically higher expression of hematopoietic and epithelial MAPs in hematopoietic (highest in DCs) and epithelial tissues (highest in mTECs), respectively, showing that MAPs lowly expressed in mTECs are highly expressed in DCs, and *vice-versa*.
(o) MAP expression on mTECs and DCs would reflect their immunogenicity. Immunogenic MAPs presented a lower expression than non-immunogenic MAPs in both types of samples.

Conclusions

BamQuery is a computational tool to evaluate the probability of MAPs presentation based on their exhaustive RNA expression, and to annotate their genomic-based features.
BamQuery showed that:

- many ERE MAPs could derive from highly expressed regions other than those previously annotated.
- published mutated neoantigens and proteasomal spliced-peptides, can be generated by other non-mutated, non-coding, non-spliced, highly expressed transcripts in normal tissues.

Moreover, it evaluates the sharing of tumor antigens' source RNAs among cancer patients. Altogether, **BamQuery** is convenient for providing immunopeptidome quantification and biological annotation in any proteogenomic study and disease context.

References & Acknowledgements

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